

2	(a)	By <u>Newton's third law</u> , both vehicles experience the <u>same magnitude of impact force</u>	B1
	(b)	As there are <u>no resultant external forces acting on the system of both the bus and car</u>, hence the <u>total momentum</u> of the system <u>is conserved</u> during the entire collision. <u>Since the initial total momentum of the two vehicles is not zero</u>, hence the final momentum of the two vehicles will also not be zero. Both vehicles will not be at rest at the same instant.	B1 B1
	(c)	$P_i = P_f$ $5m(19.4) + m(-19.4) = (5m + m)v \quad \text{M1}$ $v = 12.9 \text{ m s}^{-1} \quad \text{A1}$ Direction: to the right B1	
	(d)	The <u>car reverses its direction</u>, hence it has a <u>greater change in velocity</u> than that of the bus. Or show $\Delta v_{\text{car}} = v_f - v_i = 12.9 - (-19.4) = 32.3 \text{ m s}^{-1}$ and $\Delta v_{\text{bus}} = v_f - v_i = 12.9 - 19.4 = -6.5 \text{ m s}^{-1}$. Since both dummies have the same mass, hence <u>change in momentum</u> of the driver is <u>directly proportional</u> to the <u>change in velocity</u> experience by the driver, hence, the <u>dummy in the car experiences a greater change in momentum</u>	M1 A1
	(e)	fractional energy lost = $\frac{KE_{\text{lost}}}{KE_{\text{initial}}}$ $= \frac{KE_i - KE_f}{KE_i}$ $= \frac{\frac{1}{2}mv_i^2 - \frac{1}{2}mv_f^2}{\frac{1}{2}mv_i^2}$ $= \frac{v_i^2 - v_f^2}{v_i^2}$ $= \frac{19.4^2 - 12.9^2}{19.4^2} \quad \text{M1}$ $= 0.558 \quad \text{A1}$	

3	(a)	(i)	Diffraction is the spreading of light waves after it passes through	R1
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